

Introduction to Contract Theory and Auctions
More about revenues & Revenue Equivalence Theorem

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Expected Payment once again...

- Recall that in both auction formats, FPA and SPA, or for short I and II respectively, we have that the **interim expected payment** for each individual buyer is given by

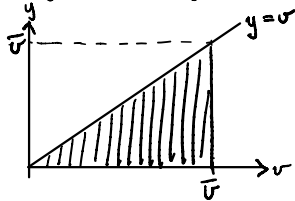
$$m^A(v) = \underbrace{G_i^{(n-1)}(v)}_{\text{Prob of Winning when value of the player is } v} \times \underbrace{E[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v]}_{\text{expected payment when winning \& value of the player is } v}$$

where $A = I, II$.

- Thus, the ex-ante payment for each individual buyer is given by

Recall that $E[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v] = \frac{1}{G_i^{(n-1)}(v)} \int_0^v y g_i^{(n-1)}(y) dy$
 Thus $\hat{m}^A(v) = \int_0^v y g_i^{(n-1)}(y) dy$

The integral is over the region $\{(v, y) : 0 \leq v \leq \bar{v}, 0 \leq y \leq v\}$. Graphically



$$\begin{aligned} E[m^A(V)] &= \int_0^{\bar{v}} m^A(v) f(v) dv = \int_0^{\bar{v}} \left(\int_0^v y g_i^{(n-1)}(y) dy \right) f(v) dv \\ &= \int_0^{\bar{v}} y g_i^{(n-1)}(y) \left(\int_y^{\bar{v}} f(v) dv \right) dy \\ &= \int_0^{\bar{v}} y g_i^{(n-1)}(y) (F(\bar{v}) - F(y)) dy \\ &= \int_0^{\bar{v}} y g_i^{(n-1)}(y) (1 - F(y)) dy \end{aligned}$$

Interchanging the order of integration

Auctioneer's Revenue once again...

- Note that the auctioneer's expected revenue in each auction format, R^A , is given by

$$\begin{aligned} E[R^A] &= n E[m^A(V)] = n \int_0^{\bar{v}} y(1 - F(y)) g_1^{(n-1)}(y) dy \\ &= \int_0^{\bar{v}} y n (1 - F(y)) g_1^{(n-1)}(y) dy \\ &= E[Y_2^{(n)}] \end{aligned}$$

But $n(1-F(y))g_1^{(n-1)}(y) = g_2^{(n)}(y)$
that is, the pdf of $Y_2^{(n)}$,
i.e., the second highest order
statistic when the sample size
is n .

$\therefore$ In both the FPA & SPA, the expected revenue of the auctioneer is the same and is equal to the expectation of the second highest order statistic with a sample size of n .

Revenues from FPA vs. Revenues of SPA

- Although we know that $E[R^I] = E[R^{II}]$, it is not the case that R^I and R^{II} are the same random variable.

Example 7.1

Recall example 6.2, two bidders with $V_1, V_2 \sim U[0, 1]$ i.i.d. and $\beta^I(v) = \frac{v}{2}$ and $\beta^{II}(v) = v$. Then, by direct calculation:

Remark: the calculations in this example only apply here!

$$R^I = \max\{\beta^I(V_1), \beta^I(V_2)\} = \beta^I(\max\{V_1, V_2\}) = \beta^I(Y_1^{(2)}) = \frac{Y_1^{(2)}}{2}$$

$$R^{II} = \min\{V_1, V_2\} = Y_2^{(2)}$$

The CDF of $R^I = Y_1^{(2)}/2 \sim 2y - y^2$ if $y \in [0, 1]$

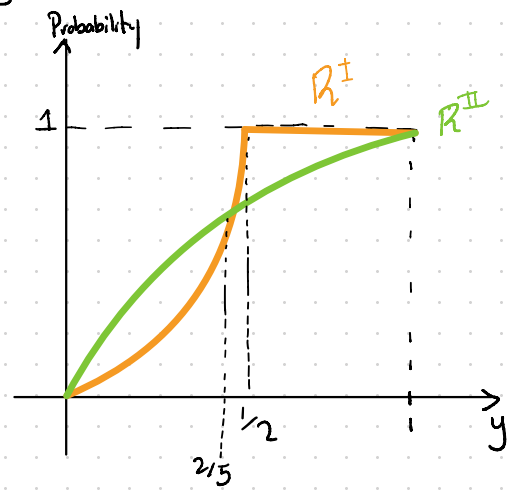
The CDF of R^I is obtained as follows
 $\Pr(R^I \leq y) = \Pr(Y_1^{(2)}/2 \leq y) = \Pr(Y_1^{(2)} \leq 2y)$
 $= (2y)^2 = 4y^2$ if $0 \leq 4y^2 \leq 1$
 $= 4y^2$ if $0 \leq y \leq \frac{1}{2}$

That is $\Pr(R^I \leq y) = \begin{cases} 0 & \text{if } y < 0 \\ 4y^2 & \text{if } 0 \leq y \leq \frac{1}{2} \\ 1 & \text{if } y \geq \frac{1}{2} \end{cases} = \min\{4y^2, 1\}$

Plotting the two CDF's

$R^I \sim \min\{4y^2, 1\}$

$R^II \sim \min\{y(2-y), 1\}$



Although $E[R^I] = E[R^{II}]$, it is clear their distributions are not the same.

- Let F_I be the distribution of R^I and F_{II} be the distribution of R^{II} .
- Example 7.1 suggests that F_{II} is a **mean-preserving spread** of F_I .
- This means that from the perspective of the auctioneer, the SPA is **riskier** than the FPA.
- Indeed, any **risk-averse auctioneer** would strictly prefer a first-price auction versus a second-price auction (assuming the bidders are risk-neutral).

Proposition 7.2

*In the SIPV auction model, the distribution of equilibrium payments in a second-price auction is a mean-preserving spread of the distribution of equilibrium payments in a first-price auction. That is, F_I **second-order stochastically dominates** F_{II} .*

Proof.



The Revenue Equivalence Theorem

- Consider a SIPV sealed-bid auction described by (q, p)
 - Recall that $q : [0, \infty)^n \rightarrow \Delta(N)$, gives the winning probability in the auction as a function of the vector of bids.
 - $p : N \times [0, \infty)^n \rightarrow \mathbb{R}^n$ gives the payment vector for each player as a function of the identity of the winner and the vector of bids.
- Let $\beta : \mathcal{V} \rightarrow [0, \infty)$ be a monotonically increasing strategy.
- Denote by $m_i(v_i) = m(q, p, \beta; v_i)$ the expected payment that buyer i with private value v_i pays in an auction method (q, p) when all the buyers implement β :

$$m_i(v_i) = \int_{v_{-i} \in \mathcal{V}^{n-1}} \sum_{i_* \in N} q_{i_*}(\beta(v_1), \dots, \beta(v_n)) p_i(i_*, \beta(v_1), \dots, \beta(v_n)) dF_{-i}(v_{-i})$$

Theorem 7.3

Let β be a symmetric and monotonically increasing equilibrium in a sealed-bid auction with independent private values satisfying the following properties:

- 1 the winner of the auction is the buyer with the highest private value.
- 2 the expected payment by a buyer with private value 0 is 0.

Then

$$m_i(v_i) = G_1^{(n-1)}(v_i) \mathbb{E}[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v_i].$$

- Property (1) is called the **efficiency condition**: an efficient auction is one in which, in equilibrium, the object is allocated to the buyer who most highly values it.
- If the seller is willing to sell the object at any nonnegative price, and since the private values of the buyers are nonnegative, an efficient auction **allocates the object with probability 1**.
- What is important about theorem 7.3: the right hand side shows that the expected payment of any bidder **does not depend on the auction method!** (Caveat: assuming the SIPV assumptions and risk-neutrality of the buyers.)

Corollary 7.4

The Revenue Equivalence Theorem In a symmetric-sealed bid auction with independent private values, let β be a symmetric and monotonically increasing equilibrium satisfying the following properties (1) and (2) from theorem 7.3. Then the seller's expected revenue is

$$\pi = n \int_{\mathcal{V}} m_i(v) f(v) dv$$

where

$$m_i(v) = G_1^{(n-1)}(v_i) \mathbb{E}[Y_1^{(n-1)} | Y_1^{(n-1)} \leq v_i].$$